evaluates the expression m + 3 using the current value of m.

m + 3
= 2 + 3
= 5

and stores the value 5 in the variable m.

m
5

The two sides of an assignment statement are separated by the symbol :=, known as assignment operator, and read as "becomes" or "is assigned". The assignment statement

v := e

is read as v "becomes" e. Note that assignment operator is not equality operator[1]. The meanings of v := e and v = e are different. Assignment does not state a mathematical equality of a variable, but changes the value of a variable. The assignment m := m + 3 does not state that m is equal to m + 3. Rather, it changes the value of the variable m to the value of the expression m + 3.

An assignment statement can change the values of multiple variables simultaneously. In that case, the number of variables on the left side should match the number of expressions on the right side. For example, if we wish to assign to three variables v1, v2 and v3, we need 3 expressions, say, el, e2, e3.

vl, v2, v3 := el, e2 , e3

The left side is a comma-separated list of variables. The right side is a comma-separated list of expressions. To execute an assignment statement, first evaluate all the expressions on the right side using the current values of the variables, and then store them in the corresponding variables on the left side.

Example 6.13. What are the values of variables m and n after the assignments in line (1) and line (3)?

1. **m, n := 2 , 5**
2. **-- m, n = ? , ?**
3. **m,n:=m+3,n-1**
4. **-- m, n = ? , ?**

The assignment in line (1) stores 2 in variable m, and 5 in variable n.

m	n
2	5

The assignment in line (3) evaluates the expressions m + 3 and n - 1 using the current values of m and n as

m + 3 , n - 1

=2+3,5-1

= 5,4

and stores the values 5 and 4 in the variables m and n, respectively.

m	n
5	4

1. **m, n := 2,5**
2. **-- m, n = 2 , 5**
3. **m, n := m + 3, n - 1**
4. **-- m, n = 2 + 3, 5-1 = 5, 4**

Values of the variables after the two assignments are shown in in line (2) and line(4).

Example 6.14. In Example 6.11, we abstracted the state of the process by two

^Unfortunately, several programming languages, including C++, use the symbol = as assignment operator, and therefore, another symbol, == as equality operator.

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